CEE6501 — Lecture 10.1

3D Frame Analysis

Learning Objectives

By the end of this lecture, you will be able to:

- derive the **pure torsion element** and its stiffness matrix
- assemble the **local 12×12 stiffness matrix** for a 3D frame element by superposition
- define the **local force** and **local displacement** vectors for a 3D frame member
- determine the **global DOFs** and numbering for a space-frame structure
- apply the **local-to-global transformation** for displacements, forces, stiffness matrices, and fixed-end-force vectors
- describe the role of the **3×3 rotation matrix** and the full **12×12 transformation matrix**
- identify the simplest **member release cases** for a 3D frame element

Agenda

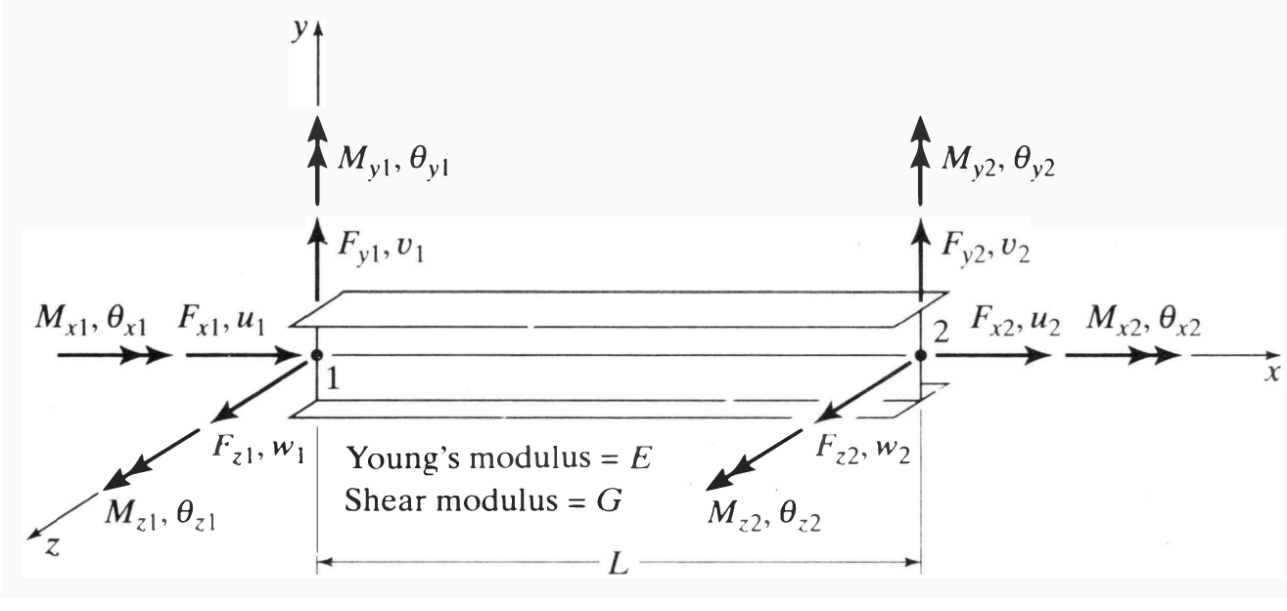
Part 1 — DSM Element Roadmap
Part 2 — Pure Torsion Member
Part 3 — Complete 3D Frame Element
Part 4 — Space Frame Global System
Part 5 — Transformation Matrix
Part 6 — Member Releases

Part 1 — DSM Element Roadmap

> *From trusses to 2D beams to 2D frames to 3D frames*

The 3D Frame Element (Where We Are Headed)

- A general 3D frame member has **6 DOFs per node** (12 per element)
  - translations: u_x, u_y, u_z
  - rotations: $\theta_x, \theta_y, \theta_z$
- This single element can represent **axial**, **torsion**, and **bending** behavior



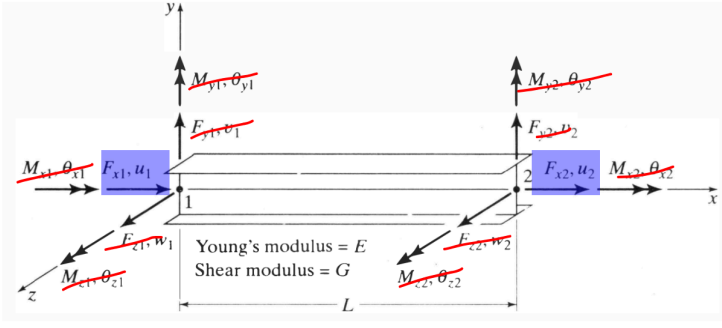
Key Idea: Superposition

You can think of the 3D frame element as a **superposition** of four behaviors:

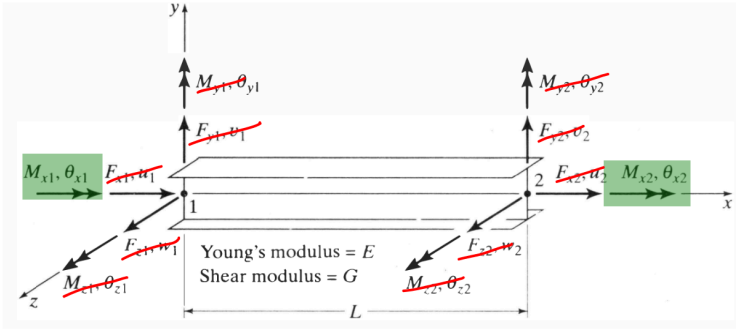
1. **Axial** deformation (truss-like)
2. **Torsion** about the member axis
3. **Bending** about one principal axis (Z-axis)
4. **Bending** about the other principal axis (Y-axis)

The Four Behaviors

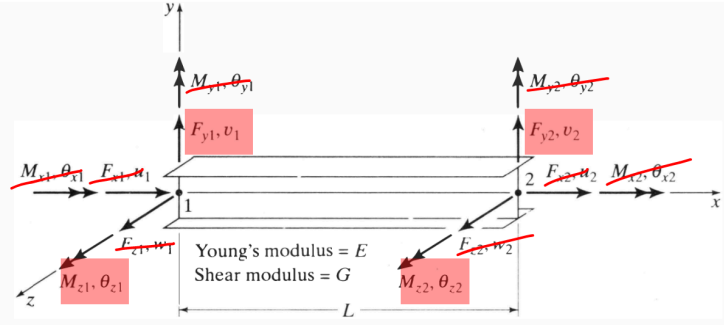
(1) Axial



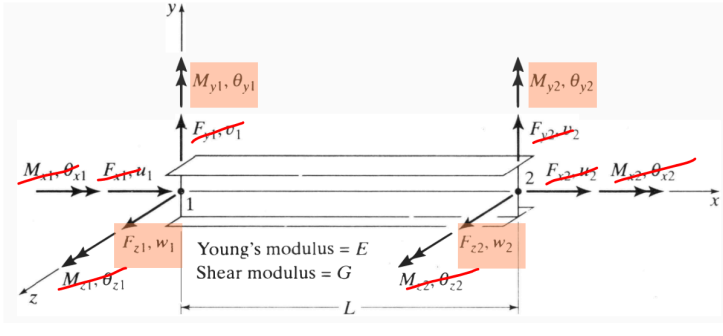
(2) Torsion



(3) Bending about Z-axis



(4) Bending about Y-axis

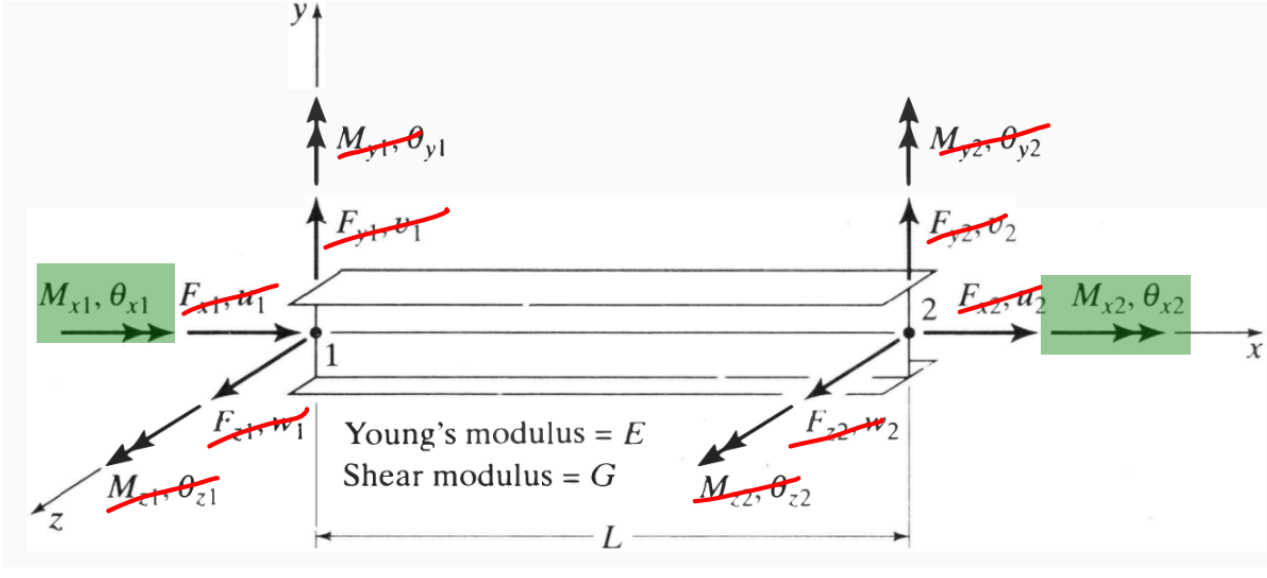


What We Have Derived So Far

- The **axial element** (case (1))
- The **2D beam element** governing planar bending behavior (cases (3) and (4))
- Once derived for one principal axis, the formulation extends directly to the other (identical mathematics, different axis)

What We Will Derive Today

- Only case (2) pure torsion element remaining
- Full DSM formulation for the generic**3D frame element**



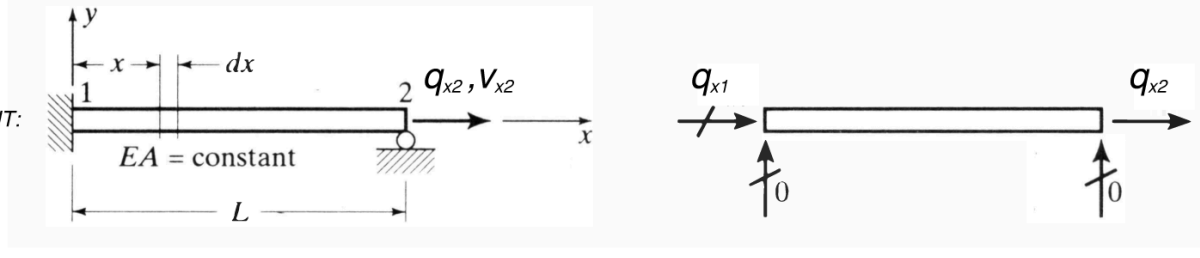
Part 2 — Pure Torsion Member

> *The missing 1D ingredient needed to build the full 3D frame element*

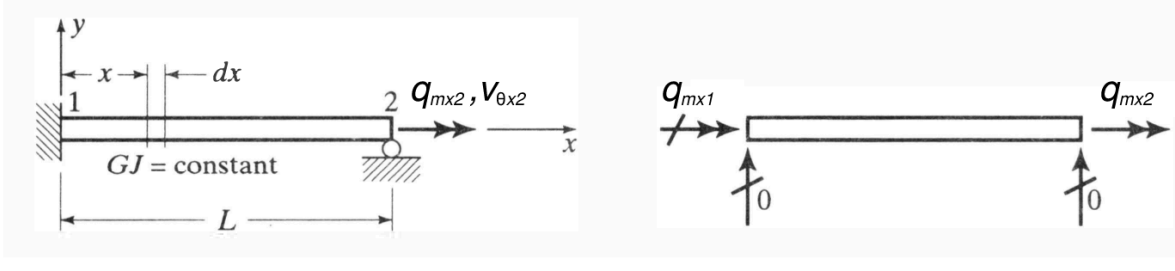
Key Analogy: Torsion Looks Just Like Axial

- Mathematically, the **pure torsion member** is directly analogous to the **axial bar element**
- The variable pair changes from
    - axial: force q_x and displacement v_x
    - torsion: torque q_{mx} and rotation $v_{\theta x}$
- So the same 2-node, 1-DOF-per-node structure appears again

AXIAL ELEMENT:



TORSION ELEMENT:



Same Mathematics, Different Physics

- In an **axial** member, the bar stretches or shortens
- In a **torsional** member, the shaft twists about its longitudinal axis
- The nodal DOFs for the torsion element are therefore the end rotations

$$v = \begin{Bmatrix} \theta_{x1} \\ \theta_{x2} \end{Bmatrix}$$

- The corresponding nodal actions are the end torques

$$q = \begin{Bmatrix} m_{x1} \\ m_{x2} \end{Bmatrix}$$

Torsional Constitutive Relation

For a prismatic shaft under pure torsion,

$$\beta = \frac{m_x}{GJ}$$

where:

- β = rate of twist
- m_x = internal torque about the member axis
- G = shear modulus (modulus of rigidity) =
- J = torsional constant of the cross section

$$G = \frac{E}{2(1+\nu)}$$

where ν is Poisson's ratio.

End Rotation From Integrating the Twist

Assuming constant GJ and constant internal torque over the element length,

$$\theta_{x2} = \int_0^L \beta \, dx = \int_0^L \frac{m_x}{GJ} \, dx = \frac{m_x L}{GJ}$$

Comparison With Axial Bar,

$$\theta_x = m_x \frac{L}{GJ}, \qquad v_x = q_x \frac{L}{EA},$$

element type	**flexibility, f**	**stiffness, k**
axial member	$\frac{L}{EA}$	$\frac{EA}{L}$
torsion member	$\frac{L}{GJ}$	$\frac{GJ}{L}$

This is exactly the same mathematical structure as the axial element, with EA replaced by GJ and v_x replaced by θ_x.

Torsional Element Stiffness Matrix

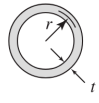
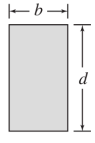
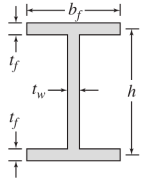
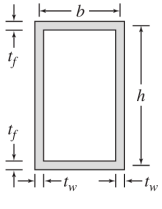
Therefore, the local torsional stiffness matrix is

$$k = \frac{GJ}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

and the element force-displacement relation becomes

$$\begin{Bmatrix} m_{x1} \\ m_{x2} \end{Bmatrix} = \frac{GJ}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{Bmatrix} \theta_{x1} \\ \theta_{x2} \end{Bmatrix}$$

List of Torsional Constants, **J**

Cross-Section	Torsion Constant
	$J = \frac{1}{2}\pi r^4$
	$J = 2\pi r^3 t$
	$J = \beta b^3 d \quad \text{for} \quad b \leq d$ $\beta = \frac{1}{3} - 0.21\frac{b}{d}\left[1 - \frac{1}{12}\left(\frac{b}{d}\right)^4\right]$
	$J = \frac{1}{3}\left(2b_f t_f^3 + h t_w^3\right)$
	$J = \frac{2b^2h^2}{b/t_f + h/t_w}$

Part 3 — Complete 3D Frame Element

Assembling axial, torsion, and two bending behaviors into one 12-DOF local element

Deriving the Local Stiffness Matrix

In principle, we could derive the full local stiffness matrix using the **unit displacement method**, just as we have done for smaller elements.

However, for a **12×12 3D frame element**, that process becomes lengthy and not especially insightful.

Instead, we take advantage of the fact that many DOFs are **uncoupled by physical behavior**, and build the full local matrix by **superposing familiar component matrices**:

- axial deformation
- torsion
- bending about the local y-axis
- bending about the local z-axis

Superposition Strategy

The complete local 3D frame element is obtained by placing the four component behaviors into the appropriate DOF locations within a single 12×12 matrix.

- the **axial block** contributes stiffness in the local x-translation DOFs
- the **torsional block** contributes stiffness in the local x-rotation DOFs
- the **bending-about-y block** couples v_z and θ_y
- the **bending-about-z block** couples v_y and θ_z

So rather than deriving one large matrix from scratch, we assemble it from these previously derived building blocks.

Local Element DOF Numbering

We use the standard **local 3D frame element** ordering, with **6 DOFs at each node** for a total of **12 element DOFs**.

Here:

- **node b** = begin node
- **node e** = end node

At each node, the local DOFs are ordered as:

- translations in the local x, y, and z directions
- rotations about the local x, y, and z axes

So the ordering is:

$$(x,\ y,\ z,\ \theta x,\ \theta y,\ \theta z)$$

for node b, followed by the same ordering for node e.

Local Force Vector

$$\mathbf{q}^T = \begin{bmatrix} q_1 & q_2 & q_3 & q_4 & q_5 & q_6 & q_7 & q_8 & q_9 & q_{10} & q_{11} & q_{12} \end{bmatrix}$$

$$= \begin{bmatrix} q_{x,b} & q_{y,b} & q_{z,b} & q_{mx,b} & q_{my,b} & q_{mz,b} & q_{x,e} & q_{y,e} & q_{z,e} & q_{mx,e} & q_{my,e} & q_{mz,e} \end{bmatrix}$$

Local Displacement Vector

$$\mathbf{v}^T = \begin{bmatrix} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & v_7 & v_8 & v_9 & v_{10} & v_{11} & v_{12} \end{bmatrix}$$

$$= \begin{bmatrix} v_{x,b} & v_{y,b} & v_{z,b} & v_{\theta x,b} & v_{\theta y,b} & v_{\theta z,b} & v_{x,e} & v_{y,e} & v_{z,e} & v_{\theta x,e} & v_{\theta y,e} & v_{\theta z,e} \end{bmatrix}$$

Local 3D Frame Stiffness Matrix

$$
\mathbf{k} = \frac{E}{L^3}\begin{bmatrix}
AL^2 & 0 & 0 & 0 & 0 & 0 & -AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & 12I_z & 0 & 0 & 0 & 6LI_z & 0 & -12I_z & 0 & 0 & 0 & 6LI_z \\
0 & 0 & 12I_y & 0 & -6LI_y & 0 & 0 & 0 & -12I_y & 0 & -6LI_y & 0 \\
0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 4L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 2L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 4L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 2L^2I_z \\
-AL^2 & 0 & 0 & 0 & 0 & 0 & AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & -12I_z & 0 & 0 & 0 & -6LI_z & 0 & 12I_z & 0 & 0 & 0 & -6LI_z \\
0 & 0 & -12I_y & 0 & 6LI_y & 0 & 0 & 0 & 12I_y & 0 & 6LI_y & 0 \\
0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 2L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 4L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 2L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 4L^2I_z
\end{bmatrix}
$$

Annotated Stiffness Matrix for All Four Components

AXIAL

$$\mathbf{k} = \frac{E}{L^3}\begin{bmatrix}
AL^2 & 0 & 0 & 0 & 0 & 0 & -AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & 12I_z & 0 & 0 & 0 & 6LI_z & 0 & -12I_z & 0 & 0 & 0 & 6LI_z \\
0 & 0 & 12I_y & 0 & -6LI_y & 0 & 0 & 0 & -12I_y & 0 & -6LI_y & 0 \\
0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 4L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 2L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 4L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 2L^2I_z \\
-AL^2 & 0 & 0 & 0 & 0 & 0 & AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & -12I_z & 0 & 0 & 0 & -6LI_z & 0 & 12I_z & 0 & 0 & 0 & -6LI_z \\
0 & 0 & -12I_y & 0 & 6LI_y & 0 & 0 & 0 & 12I_y & 0 & 6LI_y & 0 \\
0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 2L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 4L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 2L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 4L^2I_z
\end{bmatrix}$$

TORSION

$$\mathbf{k} = \frac{E}{L^3}\begin{bmatrix}
AL^2 & 0 & 0 & 0 & 0 & 0 & -AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & 12I_z & 0 & 0 & 0 & 6LI_z & 0 & -12I_z & 0 & 0 & 0 & 6LI_z \\
0 & 0 & 12I_y & 0 & -6LI_y & 0 & 0 & 0 & -12I_y & 0 & -6LI_y & 0 \\
0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 4L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 2L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 4L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 2L^2I_z \\
-AL^2 & 0 & 0 & 0 & 0 & 0 & AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & -12I_z & 0 & 0 & 0 & -6LI_z & 0 & 12I_z & 0 & 0 & 0 & -6LI_z \\
0 & 0 & -12I_y & 0 & 6LI_y & 0 & 0 & 0 & 12I_y & 0 & 6LI_y & 0 \\
0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 2L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 4L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 2L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 4L^2I_z
\end{bmatrix}$$

$$\mathbf{k} = \frac{E}{L^3}\begin{bmatrix}
AL^2 & 0 & 0 & 0 & 0 & 0 & -AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & 12I_z & 0 & 0 & 0 & 6LI_z & 0 & -12I_z & 0 & 0 & 0 & 6LI_z \\
0 & 0 & 12I_y & 0 & -6LI_y & 0 & 0 & 0 & -12I_y & 0 & -6LI_y & 0 \\
0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 4L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 2L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 4L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 2L^2I_z \\
-AL^2 & 0 & 0 & 0 & 0 & 0 & AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & -12I_z & 0 & 0 & 0 & -6LI_z & 0 & 12I_z & 0 & 0 & 0 & -6LI_z \\
0 & 0 & -12I_y & 0 & 6LI_y & 0 & 0 & 0 & 12I_y & 0 & 6LI_y & 0 \\
0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 2L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 4L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 2L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 4L^2I_z
\end{bmatrix}$$

BENDING - Z AXIS

$$\mathbf{k} = \frac{E}{L^3}\begin{bmatrix}
AL^2 & 0 & 0 & 0 & 0 & 0 & -AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & 12I_z & 0 & 0 & 0 & 6LI_z & 0 & -12I_z & 0 & 0 & 0 & 6LI_z \\
0 & 0 & 12I_y & 0 & -6LI_y & 0 & 0 & 0 & -12I_y & 0 & -6LI_y & 0 \\
0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 4L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 2L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 4L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 2L^2I_z \\
-AL^2 & 0 & 0 & 0 & 0 & 0 & AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & -12I_z & 0 & 0 & 0 & -6LI_z & 0 & 12I_z & 0 & 0 & 0 & -6LI_z \\
0 & 0 & -12I_y & 0 & 6LI_y & 0 & 0 & 0 & 12I_y & 0 & 6LI_y & 0 \\
0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 2L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 4L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 2L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 4L^2I_z
\end{bmatrix}$$

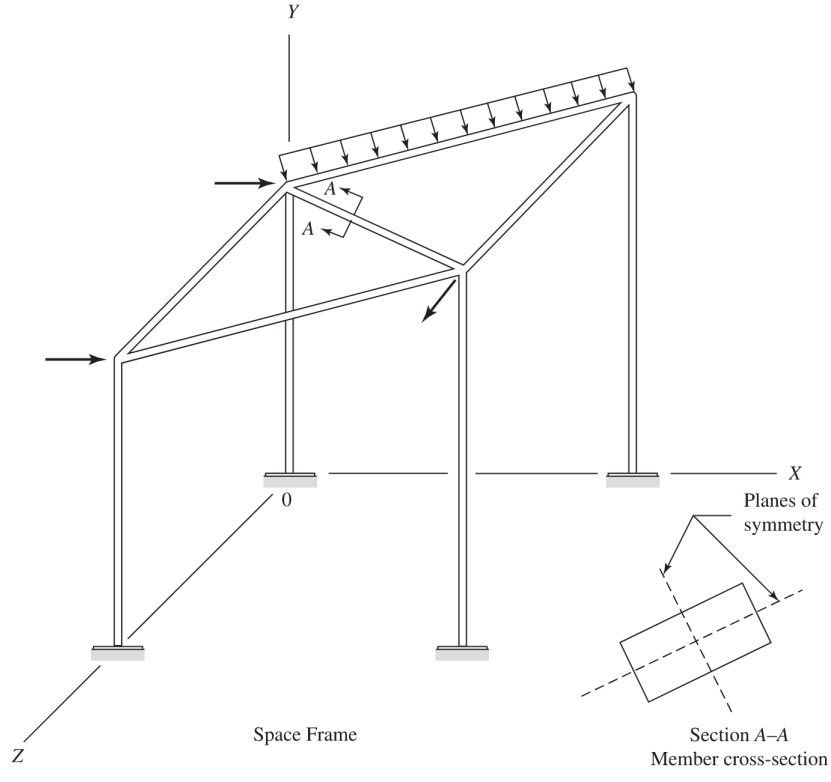
BENDING - Y AXIS

$$\mathbf{k} = \frac{E}{L^3}\begin{bmatrix}
AL^2 & 0 & 0 & 0 & 0 & 0 & -AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & 12I_z & 0 & 0 & 0 & 6LI_z & 0 & -12I_z & 0 & 0 & 0 & 6LI_z \\
0 & 0 & 12I_y & 0 & -6LI_y & 0 & 0 & 0 & -12I_y & 0 & -6LI_y & 0 \\
0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 4L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 2L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 4L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 2L^2I_z \\
-AL^2 & 0 & 0 & 0 & 0 & 0 & AL^2 & 0 & 0 & 0 & 0 & 0 \\
0 & -12I_z & 0 & 0 & 0 & -6LI_z & 0 & 12I_z & 0 & 0 & 0 & -6LI_z \\
0 & 0 & -12I_y & 0 & 6LI_y & 0 & 0 & 0 & 12I_y & 0 & 6LI_y & 0 \\
0 & 0 & 0 & -\frac{GJL^2}{E} & 0 & 0 & 0 & 0 & 0 & \frac{GJL^2}{E} & 0 & 0 \\
0 & 0 & -6LI_y & 0 & 2L^2I_y & 0 & 0 & 0 & 6LI_y & 0 & 4L^2I_y & 0 \\
0 & 6LI_z & 0 & 0 & 0 & 2L^2I_z & 0 & -6LI_z & 0 & 0 & 0 & 4L^2I_z
\end{bmatrix}$$

Part 4 — Space Frame Global System

Generic Space Frame Structure

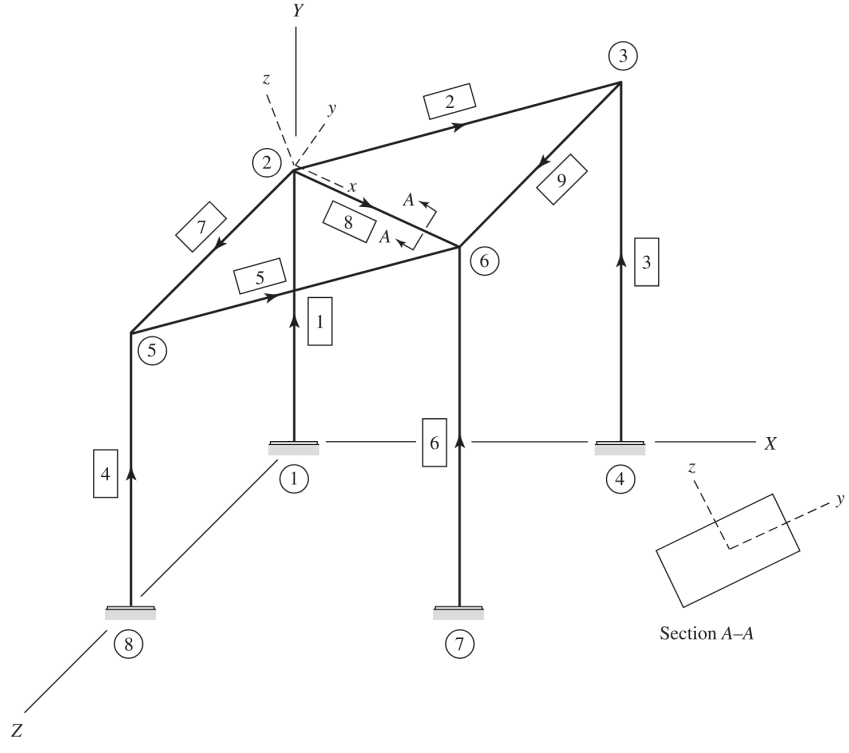
- members may be oriented in any direction in **three-dimensional space**
- connections may be **rigid**, **pinned**, or **partially restrained** depending on the modeling assumptions
- loads may act at joints or along members, and may be applied in arbitrary directions



Node and Element Numbering

The local coordinate system is defined as follows:

- the local x-axis is taken along the member centroidal axis
- the local y- and z-axes are aligned with the member cross-sectional principal directions
- the local axes are oriented to form a **right-handed** coordinate system



Global Joint Degrees of Freedom

A free joint in a space frame can:

- translate in the global X, Y, and Z directions
- rotate about the global X, Y, and Z axes

So each unconstrained joint has **6 global degrees of freedom**.

$$\text{DOFs per joint} = 6$$

Therefore, for a structure with N_J joints and N_R restrained coordinates,

$$\text{NDOF} = 6N_J - N_R$$

A stable space-frame model must include enough independent support restraints to eliminate the **6 rigid body modes** of a free 3D body: three translations and three rotations.

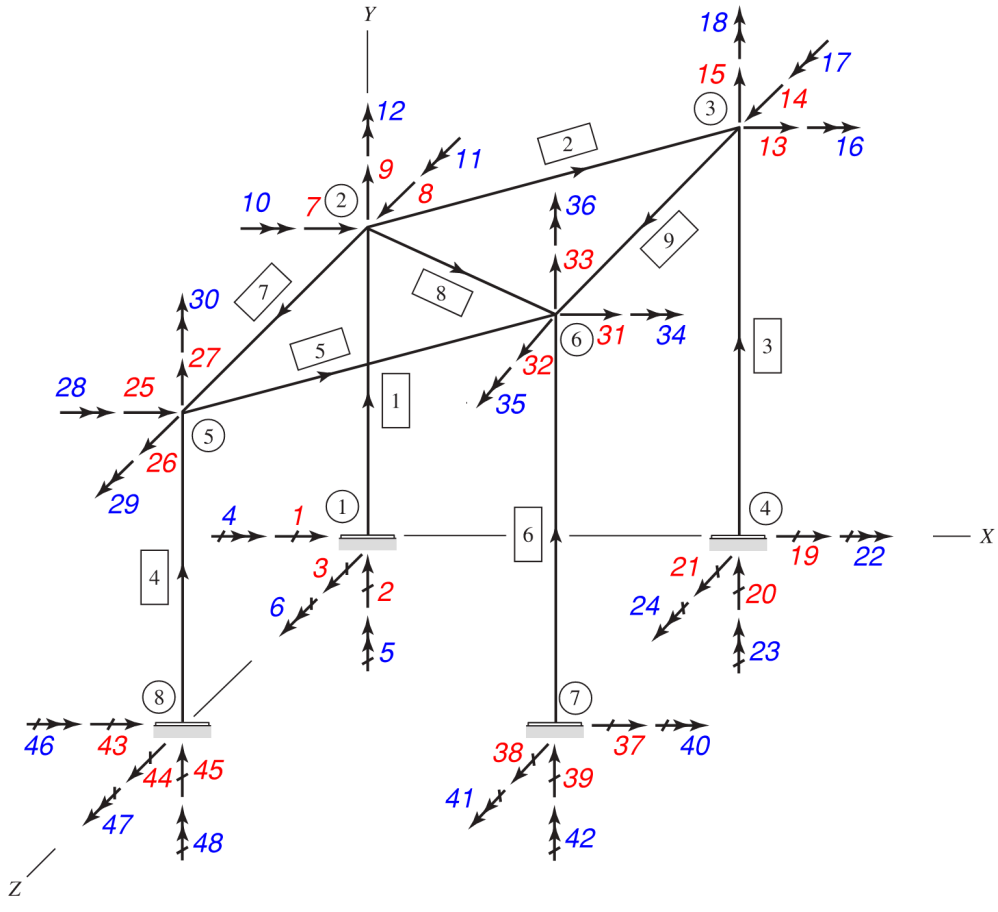
1-Based Global DOF Numbering

Global joint loads and displacements are defined with respect to a global right-handed XYZ coordinate system.

Using **1-based numbering**, the 6 DOFs associated with node n are:

$$
\begin{aligned}
6n-5 &\rightarrow X \text{ translation} \\
6n-4 &\rightarrow Y \text{ translation} \\
6n-3 &\rightarrow Z \text{ translation} \\
6n-2 &\rightarrow \text{rotation about } X \\
6n-1 &\rightarrow \text{rotation about } Y \\
6n &\rightarrow \text{rotation about } Z
\end{aligned}
$$

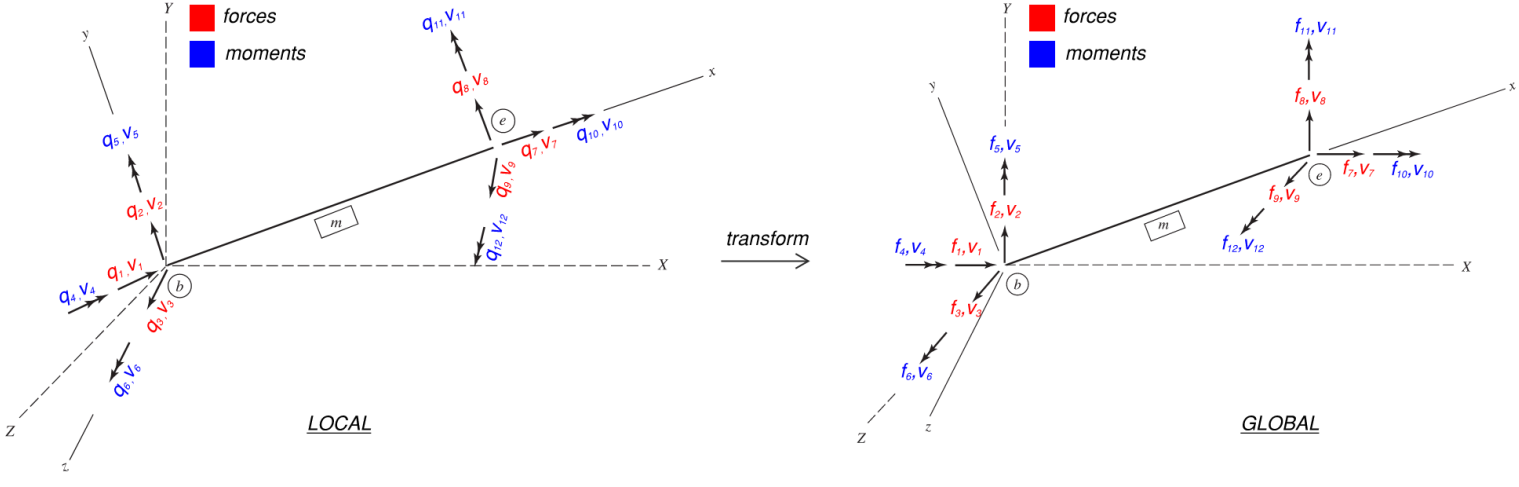
This gives a consistent way to build the global displacement vector, global force vector, and element connectivity mappings.



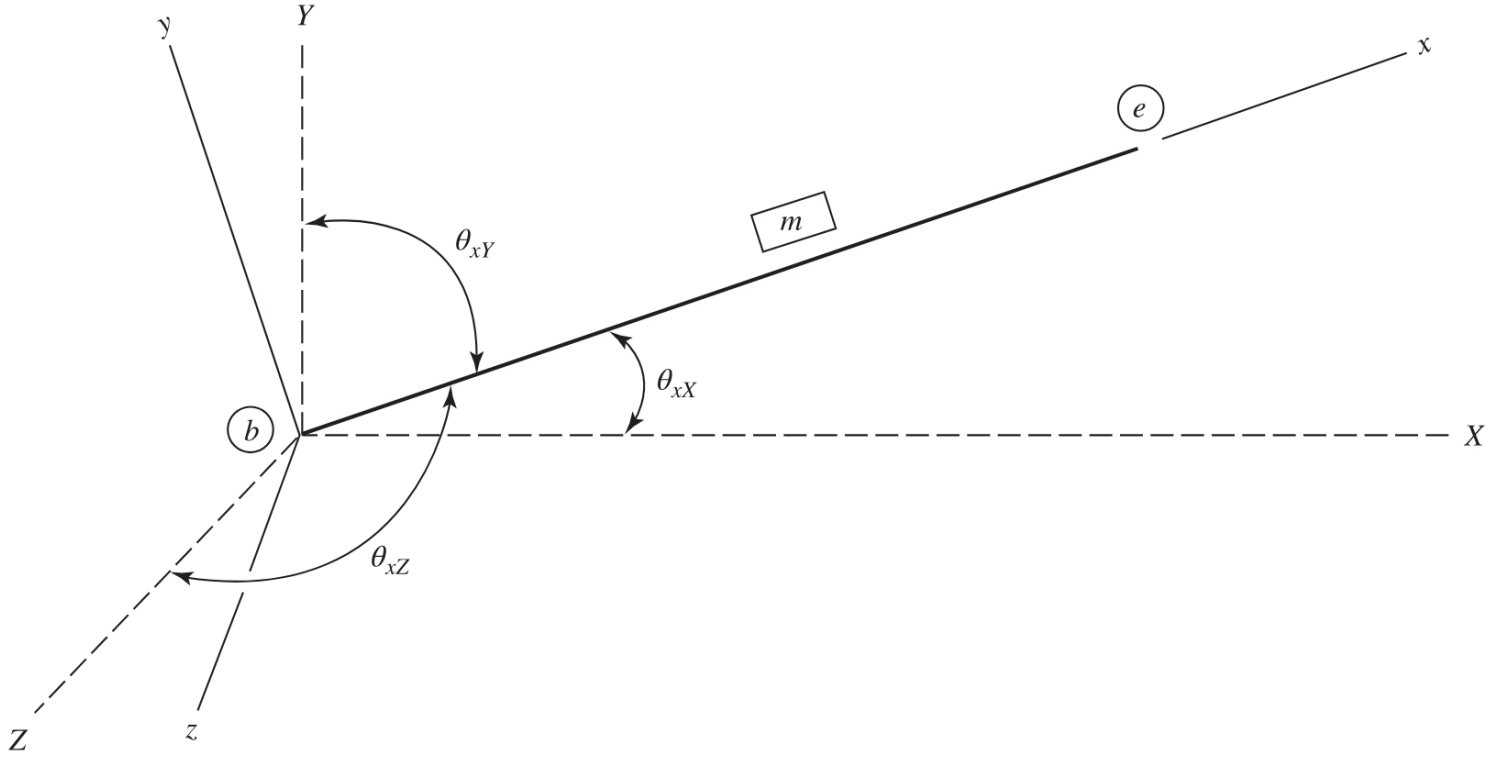
$$6n-5 \rightarrow v_{x,n}$$
$$6n-4 \rightarrow v_{y,n}$$
$$6n-3 \rightarrow v_{z,n}$$
$$6n-2 \rightarrow v_{\theta x,n}$$
$$6n-1 \rightarrow v_{\theta y,n}$$
$$6n \rightarrow v_{\theta z,n}$$

Part 5 - Transformation Matrix

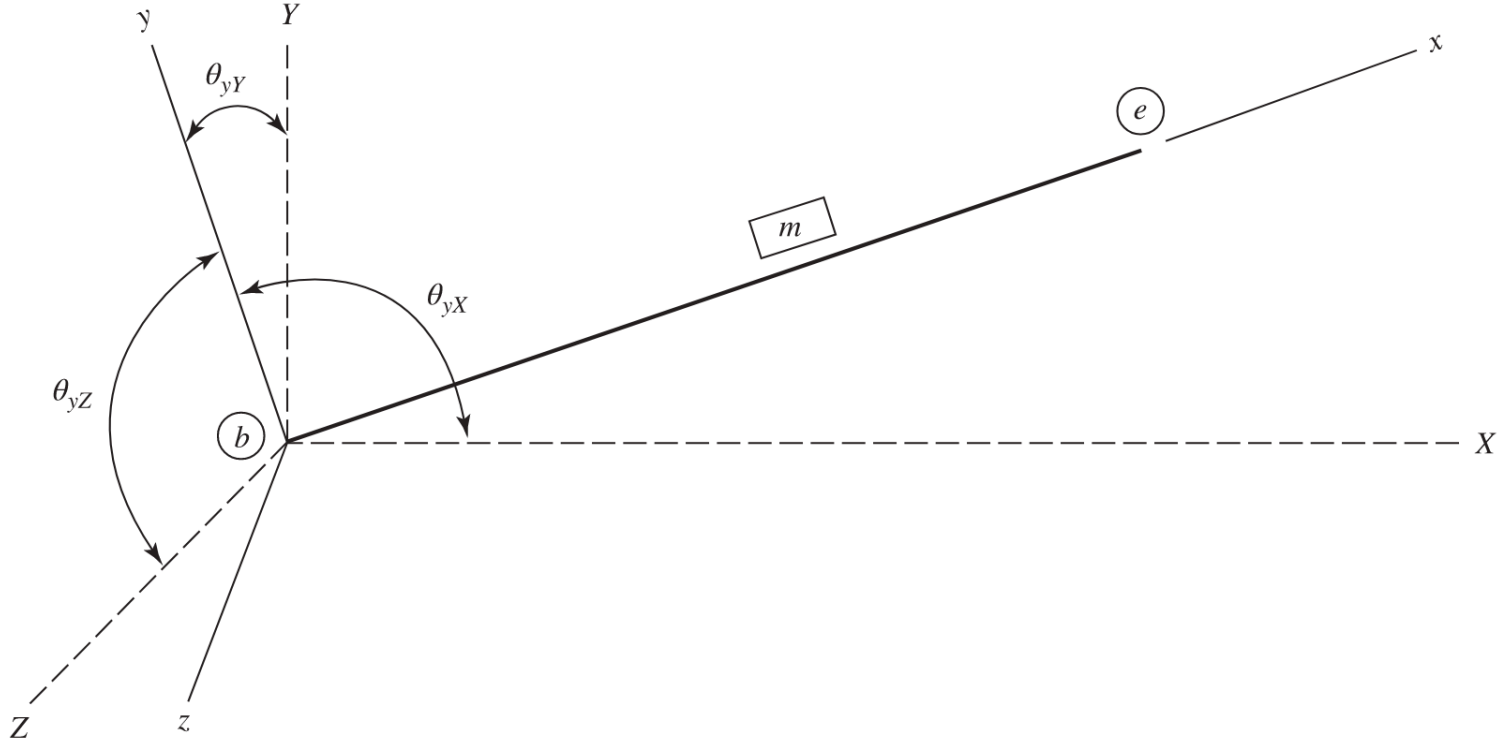
From Local to Global Coordinate System



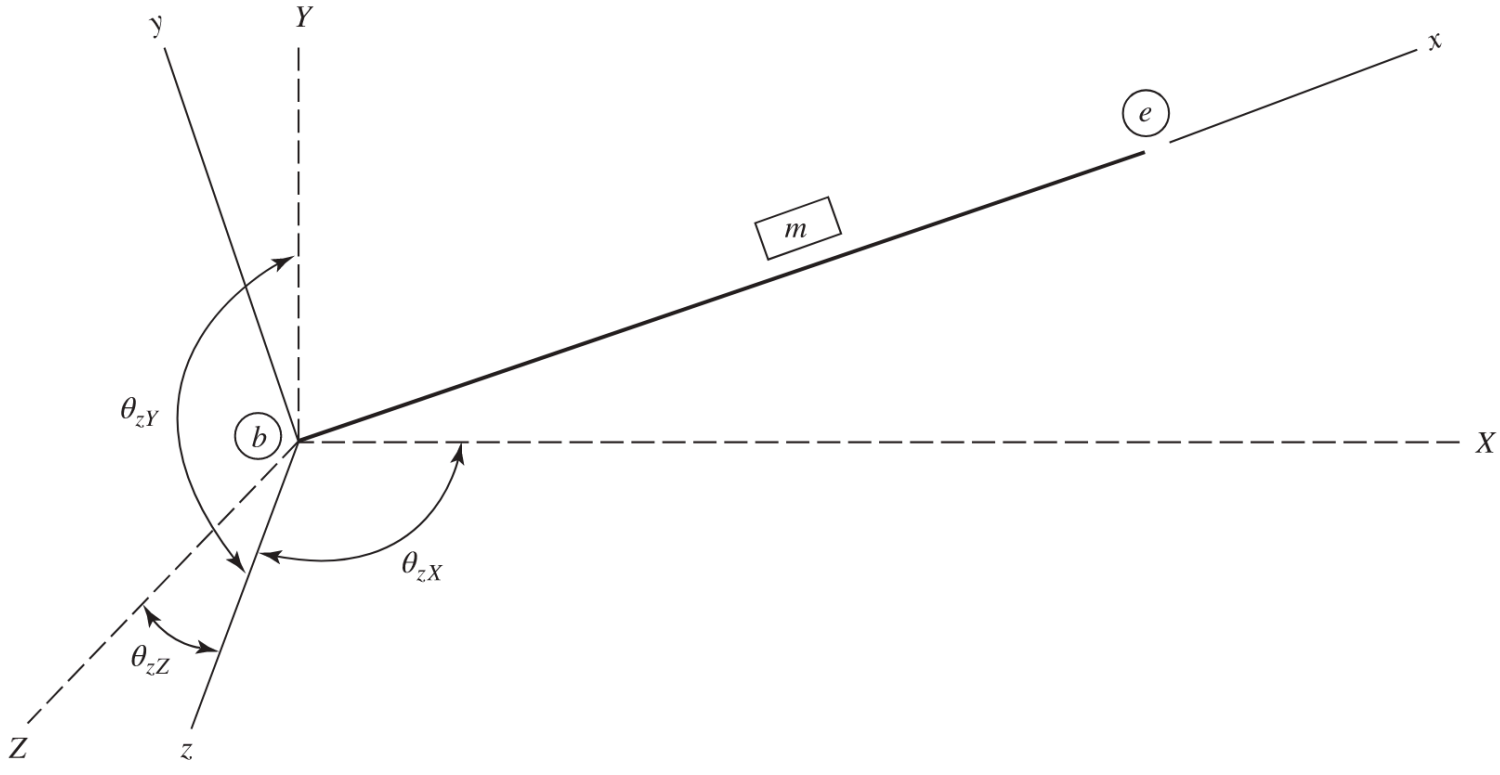
Direction Cosines, Local-X



Direction Cosines, Local-Y



Direction Cosines, Local-Z



Resolving Global Forces Into Local Components

The local force components at the start node, b, are obtained by projecting the global force vector onto the local element axes.

If the global force vector is $\mathbf{f}^T = [f_1, f_2, f_3]^T$

then the corresponding local components are

$$
\begin{aligned}
q_1 &= f_1 \cos\theta_{xX} + f_2 \cos\theta_{xY} + f_3 \cos\theta_{xZ} \\
q_2 &= f_1 \cos\theta_{yX} + f_2 \cos\theta_{yY} + f_3 \cos\theta_{yZ} \\
q_3 &= f_1 \cos\theta_{zX} + f_2 \cos\theta_{zY} + f_3 \cos\theta_{zZ}
\end{aligned}
$$

In matrix form,

$$\mathbf{q} = \mathbf{R}\mathbf{f}$$

$$\begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} r_{xX} & r_{xY} & r_{xZ} \\ r_{yX} & r_{yY} & r_{yZ} \\ r_{zX} & r_{zY} & r_{zZ} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}$$

where $\mathbf{R}$ is the 3×3 rotation matrix, and the coefficients r_{ij} are the **direction cosines** relating the local and global axes.

$$r_{ij} = \cos\theta_{ij}, \qquad i = x, y, z, \qquad j = X, Y, Z$$

Resolving Global Moments Into Local Components

The local moment components at the start node, b, are obtained in the same way, by projecting the global moment components onto the local element axes.

$$\begin{bmatrix} q_4 \\ q_5 \\ q_6 \end{bmatrix} = \begin{bmatrix} r_{xX} & r_{xY} & r_{xZ} \\ r_{yX} & r_{yY} & r_{yZ} \\ r_{zX} & r_{zY} & r_{zZ} \end{bmatrix} \begin{bmatrix} f_4 \\ f_5 \\ f_6 \end{bmatrix}$$

So the same direction-cosine matrix is used to transform both force components and moment components from global to local coordinates.

This is the same for DOFs 7 through 12.

Transformation Matrix for a Space Frame Element

Collecting the force and moment transformations for both element ends leads to the full member transformation relation between the **local** and **global** end-force vectors:

$$\mathbf{q} = \mathbf{T}\mathbf{f}$$

The transformation matrix is built from copies of the same 3×3 rotation matrix $\mathbf{R}$.

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{R} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{R} \end{bmatrix}$$

So the same 3×3 rotation block is used for:

- forces at start node, b
- moments at start node, b
- forces at end node, e
- moments at end node, e

12×12 Transformation Matrix $\mathbf{T}$

$$
\mathbf{T} = \left[\begin{array}{ccc:ccc:ccc:ccc}
r_{xX} & r_{xY} & r_{xZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
r_{yX} & r_{yY} & r_{yZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
r_{zX} & r_{zY} & r_{zZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hdashline
0 & 0 & 0 & r_{xX} & r_{xY} & r_{xZ} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & r_{yX} & r_{yY} & r_{yZ} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & r_{zX} & r_{zY} & r_{zZ} & 0 & 0 & 0 & 0 & 0 & 0 \\
\hdashline
0 & 0 & 0 & 0 & 0 & 0 & r_{xX} & r_{xY} & r_{xZ} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & r_{yX} & r_{yY} & r_{yZ} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & r_{zX} & r_{zY} & r_{zZ} & 0 & 0 & 0 \\
\hdashline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & r_{xX} & r_{xY} & r_{xZ} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & r_{yX} & r_{yY} & r_{yZ} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & r_{zX} & r_{zY} & r_{zZ}
\end{array}\right]
$$

Summary of Transformations

Once the member transformation matrix $\mathbf{T}$ is defined, we can move consistently between the **local** and **global** coordinate systems.

Global nodal displacements are transformed to local element displacements by $\mathbf{v} = \mathbf{T}\mathbf{u}$

Global element end forces are transformed to local element end forces by $\mathbf{q} = \mathbf{T}\mathbf{f}$

Global FEFs are transformed to local FEFs by $\mathbf{q}^F = \mathbf{T}\mathbf{f}^F$

Local element stiffness matrices are transformed to global element stiffness matrices by $\mathbf{K} = \mathbf{T}^T\mathbf{k}\mathbf{T}$

Local nodal displacements are transformed to global element displacements by $\mathbf{u} = \mathbf{T}^T\mathbf{v}$

Local element end forces are transformed to global element end forces by $\mathbf{f} = \mathbf{T}^T\mathbf{q}$

Local FEFs are transformed to global fixed-end-force vectors by $\mathbf{f}^F = \mathbf{T}^T\mathbf{q}^F$

Calculating the 3×3 Rotation Matrix $\mathbf{R}$

Review **Lecture 5.1** for the basic construction used for **3D truss elements**.

That same construction gives a valid 3×3 rotation matrix for a general 3D member: the member end coordinates define the local x-axis, and a local frame can then be constructed from it.

For a **3D frame element**, however, this is generally **not enough**. The member geometry defines local x, but it does not uniquely define the local y- and z-axes, since the cross section may still be rotated about the member axis.

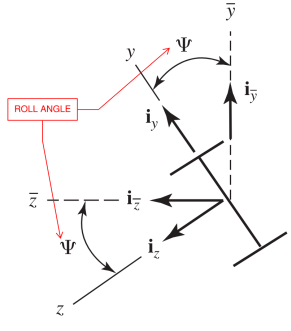
Thus, the Lecture 5.1 construction is directly sufficient only when this additional rotation is irrelevant, such as for **3D trusses** or members with **axisymmetric cross-sections** (i.e., circular)

Additional Orientation Information

For a **3D frame element**, the local coordinate system requires:

- the member axis
- the **rotation of the cross section** about that axis

In the **Lecture 5.1** construction, the cross-product procedure defines a **reference local frame**. A **roll angle** then describes the additional rotation about local x needed to obtain the actual cross-sectional orientation.



Without this extra information, the local y- and z-axes are not uniquely defined.

Complete $\mathbf{3} \times \mathbf{3}$ Rotation Matrix $\mathbf{R}$ for Frames

This form expresses the full rotation matrix in terms of the member's local x-axis direction cosines and a roll angle Ψ about that axis.

$$\begin{bmatrix} r_{xX} & r_{xY} & r_{xZ} \\ \dfrac{-r_{xX}r_{xY}\cos\Psi - r_{xZ}\sin\Psi}{\sqrt{r_{xX}^2 + r_{xZ}^2}} & \sqrt{r_{xX}^2 + r_{xZ}^2}\cos\Psi & \dfrac{-r_{xY}r_{xZ}\cos\Psi + r_{xX}\sin\Psi}{\sqrt{r_{xX}^2 + r_{xZ}^2}} \\ \dfrac{r_{xX}r_{xY}\sin\Psi - r_{xZ}\cos\Psi}{\sqrt{r_{xX}^2 + r_{xZ}^2}} & -\sqrt{r_{xX}^2 + r_{xZ}^2}\sin\Psi & \dfrac{r_{xY}r_{xZ}\sin\Psi + r_{xX}\cos\Psi}{\sqrt{r_{xX}^2 + r_{xZ}^2}} \end{bmatrix}$$

In This Course

A full 3D frame formulation requires additional information to define the cross-sectional rotation about the member axis.

Implementing this in code is beyond the scope of this course.

Therefore, we will use the same basic 3×3 rotation-matrix construction introduced in **Lecture 5.1** for **3D truss elements**.

For more details on the full derivation of $\mathbf{R}$ for a 3D frame element, see **Kassimali, 8.3**.

Part 6 - Member Releases

Types of Releases

In general, a 3D frame element may include many possible combinations of end moment releases. The modified equilibrium equations can be derived using the same basic procedure introduced earlier for **planar frame elements** in **Lecture 8**.

Because the number of possible release combinations is much larger in 3D, we will restrict attention to the simplest cases, chosen to mirror the 2D frame formulation:

- **MT = 0:** no hinges (rigid at both ends, normal 3D frame element)
- **MT = 1:** hinge at start node (**b**), all 3 moments released (ball and socket connection)
- **MT = 2:** hinge at end node (**e**), all 3 moments released (ball and socket connection)
- **MT = 3:** hinges at **both ends**, all 6 moments released (reduces to 3D truss element)

MT = 1: Stiffness Matrix

$$\mathbf{k} = \frac{E}{L^3}\begin{bmatrix} AL^2 & 0 & 0 & 0 & 0 & 0 & -AL^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3I_z & 0 & 0 & 0 & 0 & 0 & -3I_z & 0 & 0 & 0 & 3LI_z \\ 0 & 0 & 3I_y & 0 & 0 & 0 & 0 & 0 & -3I_y & 0 & -3LI_y & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -AL^2 & 0 & 0 & 0 & 0 & 0 & AL^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -3I_z & 0 & 0 & 0 & 0 & 0 & 3I_z & 0 & 0 & 0 & -3LI_z \\ 0 & 0 & -3I_y & 0 & 0 & 0 & 0 & 0 & 3I_y & 0 & 3LI_y & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3LI_y & 0 & 0 & 0 & 0 & 0 & 3LI_y & 0 & 3L^2I_y & 0 \\ 0 & 3LI_z & 0 & 0 & 0 & 0 & 0 & -3LI_z & 0 & 0 & 0 & 3L^2I_z \end{bmatrix}$$

MT = 1: Fixed End Force Vector

$$\mathbf{q}^F_{(MT=1)} = \begin{bmatrix} f^F_{x,b} \\ f^F_{y,b} - \dfrac{3}{2L} f^F_{mz,b} \\ f^F_{z,b} + \dfrac{3}{2L} f^F_{my,b} \\ \boxed{0} \\ \boxed{0} \\ \boxed{0} \\ f^F_{x,e} \\ f^F_{y,e} + \dfrac{3}{2L} f^F_{mz,b} \\ f^F_{z,e} - \dfrac{3}{2L} f^F_{my,b} \\ f^F_{tx,b} + f^F_{tx,e} \\ f^F_{my,e} - \dfrac{1}{2} f^F_{my,b} \\ f^F_{mz,e} - \dfrac{1}{2} f^F_{mz,b} \end{bmatrix}$$

MT = 2: Stiffness Matrix

$$\mathbf{k} = \frac{E}{L^3}\begin{bmatrix} AL^2 & 0 & 0 & 0 & 0 & 0 & -AL^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3I_z & 0 & 0 & 0 & 3LI_z & 0 & -3I_z & 0 & 0 & 0 & 0 \\ 0 & 0 & 3I_y & 0 & -3LI_y & 0 & 0 & 0 & -3I_y & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3LI_y & 0 & 3L^2I_y & 0 & 0 & 0 & 3LI_y & 0 & 0 & 0 \\ 0 & 3LI_z & 0 & 0 & 0 & 3L^2I_z & 0 & -3LI_z & 0 & 0 & 0 & 0 \\ -AL^2 & 0 & 0 & 0 & 0 & 0 & AL^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -3I_z & 0 & 0 & 0 & -3LI_z & 0 & 3I_z & 0 & 0 & 0 & 0 \\ 0 & 0 & -3I_y & 0 & 3LI_y & 0 & 0 & 0 & 3I_y & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

MT = 2: Fixed End Force Vector

$$\mathbf{q}^F_{(MT=2)} = \begin{bmatrix} f^F_{x,b} \\ f^F_{y,b} + \dfrac{3}{2L} f^F_{mz,e} \\ f^F_{z,b} - \dfrac{3}{2L} f^F_{my,e} \\ f^F_{tx,b} + f^F_{tx,e} \\ f^F_{my,b} - \dfrac{1}{2} f^F_{my,e} \\ f^F_{mz,b} - \dfrac{1}{2} f^F_{mz,e} \\ f^F_{x,e} \\ f^F_{y,e} - \dfrac{3}{2L} f^F_{mz,e} \\ f^F_{z,e} + \dfrac{3}{2L} f^F_{my,e} \\ \boxed{0} \\ \boxed{0} \\ \boxed{0} \end{bmatrix}$$

MT = 3: Fixed End Force Vector

$$\mathbf{k} = \frac{EA}{L}\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

MT = 3: Fixed End Force Vector

$$\mathbf{q}^F_{(MT=3)} = \begin{bmatrix} f^F_{x,b} \\ f^F_{y,b} - \frac{1}{L}\left(f^F_{mz,b} + f^F_{mz,e}\right) \\ f^F_{z,b} + \frac{1}{L}\left(f^F_{my,b} + f^F_{my,e}\right) \\ \boxed{0} \\ \boxed{0} \\ \boxed{0} \\ f^F_{x,e} \\ f^F_{y,e} + \frac{1}{L}\left(f^F_{mz,b} + f^F_{mz,e}\right) \\ f^F_{z,e} - \frac{1}{L}\left(f^F_{my,b} + f^F_{my,e}\right) \\ \boxed{0} \\ \boxed{0} \\ \boxed{0} \end{bmatrix}$$

Wrap-Up

In this lecture, we:

- completed the final missing 1D ingredient: the **pure torsion element**
- assembled the full **12-DOF local stiffness matrix** for a 3D frame element
- introduced **space-frame DOF numbering** and global-system setup
- developed the **local-to-global transformation** using the rotation matrix
- introduced the simplest **3D member release cases**: $MT = 0, 1, 2, 3$

For this course, we will use the **Lecture 5.1** construction for the 3×3 rotation matrix and not pursue the full roll-angle-based frame orientation formulation.